

01 GET /alarms

RULES ["compatibility aliases derive only from canonical fields", "source R03 requires alarm_type OOS, rule_code R03_CONSEC, and value null", "source SUMMARY requires rule_code SUMMARY_OOC", "source TRACE requires rule_code TRACE_OOS"]

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        "required": false,
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          "type": "string"
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          "schema": {
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            "required": false,
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                    "fields": {
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                        "required": false,
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                          "type": "string"
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```

02 GET /trace

RULES []

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        "required": true,
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        "required": true,
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                "required": true,
                "schema": {
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                  "type": "integer"
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}
```

03 GET /parameters

RULES []

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  "request": {
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    "header": {},
    "path": {},
    "query": {},
    "responses": {
      "200": {
        "schema": {
          "item
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04 POST /documents/search

RULES []

{ "request": { "body": { "additional_properties": false, "fields": { "model_code": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "query": { "nullable": false, "required": true, "schema": { "max_length": 1000, "min_length": 1, "type": "string" } }, "top_k": { "nullable": false, "required": false, "schema": { "default": 4, "maximum": 10.0, "minimum": 1.0, "type": "integer" } }, "type": "object" }, "header": {}, "path": {}, "query": {}, "responses": { "200": { "schema": { "items": { "additional_properties": false, "fields": { "chunk_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "content": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "doc_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "document_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "model_code": { "nullable": true, "required": false, "schema": { "min_length": 1, "type": "string" } }, "score": { "nullable": false, "required": true, "schema": { "maximum": 1.0, "minimum": -1.0, "type": "number" } }, "section": { "nullable": true, "required": false, "schema": { "type": "string" } }, "title": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "type": "array", "shape": "array" }, "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "items": { "type": "union", "variants": [{ "type": "string" }, { "type": "integer" }] }, "type": "array" }, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object" }, "type": "array" } }, "type": "object", "shape": "object" } }

05 GET /relations/chambers/{chamber_id}

RULES ["node_count equals len(nodes)", "relationship_count equals len(relationships)"]
{ "request": { "body": null, "header": {}, "path": { "chamber_id": { "nullable": false, "required": true, "schema": { "type": "string" } }, "query": {}, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "graph_revision": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "nodes": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "business_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "display_name": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "label": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "properties": { "nullable": false, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" } }, "type": "object", "type": "array" }, "relationships": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": false, "fields": { "id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "source": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "target": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "type": "array" }, "root_node_id": { "nullable": false, "required": true, "schema": { "min_length": 1, "type": "string" } }, "type": "object", "shape": "object", "422": { "schema": { "additional_properties": true, "fields": { "detail": { "nullable": false, "required": false, "schema": { "items": { "additional_properties": true, "fields": { "loc": { "nullable": false, "required": true, "schema": { "items": { "type": "union", "variants": [{ "type": "string" }, { "type": "integer" }] }, "type": "array" }, "msg": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object", "type": "array" } }, "type": "object", "shape": "object" } }

06 GET /agent/runs

RULES []

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RULES [{"fault_code": "ground_truth_fault_code are forbidden", "metrology_alarm_result is forbidden", "request": {"body": {"additional_properties": false, "fields": {"question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "header": {"path": {}, "query": {}}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {"answer": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "confidence": {"nullable": true, "required": true, "schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "evidence": {"nullable": true, "required": true, "schema": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}, "evidence_items": {"nullable": false, "required": true, "schema": {"items": {"discriminator": "type", "type": "discriminated_union"}, "variants": {"ALARM": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"type": "string"}}, "DOCUMENT": {"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "string"}}, "GRAPH": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "graph_revision": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "relation_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "string"}}, "METROLOGY": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "string"}}, "TRACE": {"additional_properties": false, "fields": {"excerpt": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "source_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": {"type": "string"}}, "type": {"type": "string"}}, "array": {"limit": {"nullable": false, "required": true, "schema": {"type": "string"}}, "limitations": {"nullable": false, "required": true, "schema": {"items": {"type": "string"}, "type": "array"}}, "predicted_fault_code": {"nullable": true, "required": true, "schema": {"enum": ["FOC", "MFD", "OTH", "RFM", "TMD"], "type": "string"}}, "recommended_action": {"nullable": true, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "title": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "tools": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"name": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "result": {"nullable": false, "required": true, "schema": {"min_length":

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08 GET /approvals
RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields n
ull", "only EQP_HOLD actions have approvals"]
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09 POST /approvals/{approval_id}/decision
RULES ["APPROVED or REJECTED requires decided_by and decided_at", "PENDING requires all decision fields n
ull", "only EQP_HOLD actions have approvals"]
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11 GET /dataset/bounds
RULES []
deferred inventory
12 GET /dashboard/summary
RULES []
deferred inventory
13 GET /alarms/{source}/{alarm_id}
RULES []
deferred inventory
14 GET /alarms/paged
RULES []
deferred inventory
15 GET /traces/catalog
RULES []
deferred inventory
16 POST /traces/search
RULES []
deferred inventory
17 GET /relations/equipment/{equipment_id}
RULES []
deferred inventory
18 GET /documents/{document_id}
RULES []
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19 POST /agent/runs
RULES []
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[illegible]

[illegible]

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25 GET /actions/{action_id}
RULES []
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28 POST /analytics/query
RULES []
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29 POST /analytics/graph-query
RULES []
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.0, "type": "integer"}}, "question": {"nullable": false, "required": true, "schema": {"type": "string"}},
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lable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, "rows": {"nullable": fal
se, "required": true, "schema": {"items": {"additional_properties": true, "fields": {}, "type": "object"}
, "type": "array"}}, "type": "object"}, "shape": "object", "422": {"schema": {"additional_properties":
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, "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable": false, "
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type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "shape": "object"}}}}
30 POST /analytics/validate
RULES []
{"request": {"body": {"additional_properties": false, "fields": {"sql": {"nullable": false, "required": t
```

```
rue,"schema":{"max_length": 20000,"min_length": 1,"type": "string"}}},"type": "object"},"header": {
  }, "path": {}, "query": {}, "responses": {"200": {"schema": {"additional_properties": false, "fields": {
    "checks": {"nullable": true, "required": false, "schema": {"items": {"additional_properties": false, "fi
    e lds": {"key": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "labe
    l": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "ok": {"nullabl
    e": false, "required": true, "schema": {"type": "boolean"}}}, "type": "object"}, "type": "array"}}, "norm
    alized_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "reason": {"nullable":
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    erties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additi
    onal_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type"
    : "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg": {"nullable":
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    chema": {"type": "string"}}}, "type": "object"}, "type": "array"}}}, "type": "object"}, "shape": "object"
  }}}}
```

31 GET /analytics/history

RULES []

```
{ "request": {"body": null, "header": {}, "path": {}, "query": {"date_from": {"nullable": true, "required"
: false, "schema": {"format": "date", "type": "string"}}, "date_to": {"nullable": true, "required": false
, "schema": {"format": "date", "type": "string"}}, "is_rejected": {"nullable": true, "required": false, "
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ean"}}, "page": {"nullable": true, "required": false, "schema": {"default": 1, "minimum": 1, "type": "in
teger"}}, "size": {"nullable": false, "required": false, "schema": {"default": 20, "maximum": 100, "minim
um": 1, "type": "integer"}}}, "responses": {"200": {"schema": {"additional_properties": false, "fields":
{"items": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fi
elds": {"asked_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "stri
ng"}}, "error_msg": {"nullable": true, "required": false, "schema": {"type": "string"}}, "generated_sql":
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ma": {"type": "boolean"}}, "latency_ms": {"nullable": true, "required": false, "schema": {"minimum": 0.0,
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" type": "integer"}}, "outcome": {"nullable": false, "required": true, "schema": {"enum": ["DB_ERROR", "P
OLICY_REJECTED", "SUCCESS", "VALIDATION_FAILED"], "type": "string"}}, "question": {"nullable": false, "re
quired": true, "schema": {"type": "string"}}, "reject_reason": {"nullable": true, "required": false, "sch
ema": {"type": "string"}}, "row_cnt": {"nullable": true, "required": false, "schema": {"minimum": 0.0, "t
ype": "integer"}}}, "type": "object"}, "type": "array"}}, "page": {"nullable": false, "required": true, "
schema": {"minimum": 1.0, "type": "integer"}, "size": {"nullable": false, "required": true, "schema": {"
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s": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"i
tems": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, "type": "array"}}, "msg"
: {"nullable": false, "required": true, "schema": {"type": "string"}}, "type": {"nullable": false, "requi
red": true, "schema": {"type": "string"}}, "type": "object"}, "type": "array"}}, "type": "object"}, "sh
ape": "object"}}}
```

32 GET /analytics/evaluations

RULES []

```
{ "request": {"body": null, "header": {}, "path": {}, "query": {"latest": {"nullable": false, "required":
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"schema": {"default": 20, "maximum": 100, "minimum": 1, "type": "integer"}}}, "responses": {"200": {"sch
ema": {"additional_properties": false, "fields": {"items": {"nullable": false, "required": true, "schema"
: {"items": {"additional_properties": false, "fields": {"accuracy": {"nullable": false, "required": true,
"schema": {"maximum": 1.0, "minimum": 0.0, "type": "number"}}, "correct": {"nullable": false, "required"
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true, "schema": {"minimum": 0.0, "type": "integer"}}, "defense_total": {"nullable": false, "required": t
rue, "schema": {"minimum": 0.0, "type": "integer"}}, "executed_at": {"nullable": false, "required": true,
"schema": {"format": "date-time", "type": "string"}}, "items": {"nullable": false, "required": true, "sc
hema": {"items": {"additional_properties": false, "fields": {"actual_result": {"nullable": true, "require
d": false, "schema": {"type": "any"}}, "actual_visualization": {"nullable": true, "required": false, "sch
ema": {"additional_properties": false, "fields": {"chart_type": {"nullable": false, "required": true, "sc
hema": {"enum": ["bar", "histogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "requi
red": false, "schema": {"type": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type":
"string"}}, "type": "object"}}, "attempt_count": {"nullable": false, "required": true, "schema": {"mini
mum": 0.0, "type": "integer"}}, "case_id": {"nullable": false, "required": true, "schema": {"min_length":
1, "type": "string"}}, "case_type": {"nullable": false, "required": true, "schema": {"enum": ["DEFENSE",
"GOLD"], "type": "string"}}, "expected_result": {"nullable": true, "required": false, "schema": {"type":
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es": false, "fields": {"chart_type": {"nullable": false, "required": true, "schema": {"enum": ["bar", "hi
stogram", "line", "table"], "type": "string"}}, "x": {"nullable": true, "required": false, "schema": {"ty
pe": "string"}}, "y": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "obje
ct"}}, "generated_sql": {"nullable": true, "required": false, "schema": {"type": "string"}}, "latency_ms"
: {"nullable": true, "required": false, "schema": {"minimum": 0.0, "type": "integer"}}, "passed": {"nulla
ble": false, "required": true, "schema": {"type": "boolean"}}, "question": {"nullable": true, "required":

```

```
false, "schema": {"type": "string"}}, "reason": {"nullable": true, "required": false, "schema": {"type": "string"}}, "type": "object"}, {"type": "array"}, {"model": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"prompt_version": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"provider": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, {"temperature": {"nullable": false, "required": true, "schema": {"type": "number"}}, {"total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"type": "object"}, {"type": "array"}, {"page": {"nullable": false, "required": true, "schema": {"minimum": 1.0, "type": "integer"}}, {"size": {"nullable": false, "required": true, "schema": {"maximum": 100.0, "minimum": 1.0, "type": "integer"}}, {"total": {"nullable": false, "required": true, "schema": {"minimum": 0.0, "type": "integer"}}, {"type": "object"}, {"shape": "object"}, {"422": {"schema": {"additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, {"type": "array"}}, {"msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"type": "array"}}, {"type": "object"}, {"shape": "object"}}}
```

33 GET /audit-logs/paged

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": { "actor_type": { "nullable": true, "required": false, "schema": { "type": "string" } }, "date_from": { "nullable": true, "required": false, "schema": { "format": "date", "type": "string" } }, "date_to": { "nullable": true, "required": false, "schema": { "format": "date", "type": "string" } }, "entity_id": { "nullable": true, "required": false, "schema": { "type": "string" } }, "event_type": { "nullable": true, "required": false, "schema": { "type": "string" } }, "page": { "nullable": false, "required": false, "schema": { "default": 1, "minimum": 1, "type": "integer" } }, "size": { "nullable": false, "required": false, "schema": { "default": 20, "maximum": 100, "minimum": 1, "type": "integer" } } }, "responses": { "200": { "schema": { "additional_properties": true, "fields": { "event_type_counts": { "nullable": false, "required": true, "schema": { "additional_properties": { "type": "integer" }, "fields": {}, "type": "object" }, "event_types": { "nullable": false, "required": true, "schema": { "items": { "type": "string", "type": "array" }, "items": { "nullable": false, "required": true, "schema": { "items": { "additional_properties": true, "fields": { "actor": { "nullable": false, "required": true, "schema": { "type": "string" } }, "actor_id": { "nullable": true, "required": true, "schema": { "type": "string" } }, "actor_type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "after": { "nullable": true, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" }, "at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } }, "audit_id": { "nullable": false, "required": true, "schema": { "type": "integer" } }, "before": { "nullable": true, "required": true, "schema": { "additional_properties": true, "fields": {}, "type": "object" }, "detail": { "nullable": true, "required": true, "schema": { "type": "string" } }, "entity": { "nullable": false, "required": true, "schema": { "type": "string" } }, "entity_id": { "nullable": false, "required": true, "schema": { "type": "string" } }, "entity_type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "event": { "nullable": false, "required": true, "schema": { "type": "string" } }, "event_type": { "nullable": false, "required": true, "schema": { "type": "string" } }, "occurred_at": { "nullable": false, "required": true, "schema": { "format": "date-time", "type": "string" } }, "type": "object", "type": "array", "page": { "nullable": false, "required": true, "schema": { "type": "integer" }, "size": { "nullable": false, "required": true, "schema": { "type": "integer" }, "total": { "nullable": false, "required": true, "schema": { "type": "integer" }, "type": "object", "shape": "object", "422": { "schema": { "additional_properties": true, "fields": {"detail": {"nullable": false, "required": false, "schema": {"items": {"additional_properties": true, "fields": {"loc": {"nullable": false, "required": true, "schema": {"items": {"type": "union", "variants": [{"type": "string"}, {"type": "integer"}]}, {"type": "array"}}, {"msg": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": {"nullable": false, "required": true, "schema": {"type": "string"}}, {"type": "object"}, {"type": "array"}}, {"type": "object"}, {"shape": "object"}}}
```

34 GET /health

RULES []

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {}, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "status": { "nullable": false, "required": true, "schema": { "type": "string" } }, "type": "object", "shape": "object" } } }
```

35 GET /health/ready

RULES ["all checks PASS yields status READY and HTTP 200", "any check FAIL yields status NOT_READY and HTTP 503"]

```
{ "request": { "body": null, "header": {}, "path": {}, "query": {}, "responses": { "200": { "schema": { "additional_properties": false, "fields": { "checks": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "kafka": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "latency_ms": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer" }, "reason_code": { "nullable": true, "required": true, "schema": { "enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string" }, "status": { "nullable": false, "required": true, "schema": { "enum": ["FAIL", "PASS"], "type": "string" } }, "type": "object", "n8n": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "latency_ms": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer" }, "reason_code": { "nullable": true, "required": true, "schema": { "enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string" }, "status": { "nullable": false, "required": true, "schema": { "enum": ["FAIL", "PASS"], "type": "string" } }, "type": "object", "neo4j": { "nullable": false, "required": true, "schema": { "additional_properties": false, "fields": { "latency_ms": { "nullable": false, "required": true, "schema": { "minimum": 0.0, "type": "integer" }, "reason_code": { "nullable": true, "required": true, "sc
```



```
required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "complete_d_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "start_d_at": {"nullable": true, "required": true, "schema": {"format": "date-time", "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

ActionItem

```
{"additional_properties": false, "fields": {"action_code": {"nullable": false, "required": true, "schema": {"enum": ["EQP_HOLD", "MONITORING", "WARNING"], "type": "string"}}, "action_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "approval_status": {"nullable": true, "required": true, "schema": {"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}}, "chamber": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "chamber_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "created_at": {"nullable": false, "required": true, "schema": {"format": "date-time", "type": "string"}}, "created_by_agent_run_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "deliveries": {"nullable": false, "required": true, "schema": {"items": {"additional_properties": false, "fields": {"channel": {"nullable": false, "required": true, "schema": {"enum": ["EMAIL", "MES"], "type": "string"}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["BLOCKED", "CANCELED", "FAILED", "SENDING", "SENT", "UNKNOWN", "WAITING"], "type": "string"}}, "type": "object"}, "type": "array"}}, "equipment": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "equipment_id": {"nullable": true, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "lot_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "reason": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "type": "object"}
```

AgentAskCompatibilityEvidence

```
{"additional_properties": false, "fields": {"chunk_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "doc_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "document_id": {"nullable": false, "required": true, "schema": {"min_length": 1, "type": "string"}}, "section": {"nullable": true, "required": true, "schema": {"type": "string"}}, "type": "object"}
```

AgentAskRequest

```
{"additional_properties": false, "fields": {"question": {"nullable": false, "required": true, "schema": {"max_length": 1000, "min_length": 1, "type": "string"}}, "type": "object"}
```

AgentAskResponse

```
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AgentRunAcceptedResponse

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AgentRunActionItem

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AgentRunApprovalItem

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AgentRunDetailResponse

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AgentRunItem
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AlarmSource

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AnalysisQueryRequest

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AnalysisQueryResponse

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ApprovalDecisionRequest

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ApprovalItem

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AskDocumentEvidenceAlias

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AskToolItem

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AuditLogItem

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AuditLogPageResponse

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AutoToolCallItem

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ChamberGraphResponse

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    "required": true,
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    "fields": {
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        "required": true,
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        "type": "string"
      },
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        "required": true,
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        "required": true,
        "schema": {}
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    "type": "object"
  },
  "rules": [
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    "FAILED requires error_code non-null and permits provider_message_id null"
  ],
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ChartType
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ChatToolCallItem
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CrossCheck
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DeliveryChannel
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DeliveryResult
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DocumentDetailResponse
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ErrorCode
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FaultHypothesis
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GraphContext
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GraphQueryResponse
```

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  "fields": {
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      "required": true,
      "schema": {
        "items": {
          "type": "string",
          "type": "array"
        },
        "error_msg": {
          "nullable": true,
          "required": false,
          "schema": {
            "type": "string"
          }
        },
        "generated_cypher": {
          "nullable": true,
          "required": false,
          "schema": {
            "type": "string"
          }
        },
        "is_rejected": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "boolean"
          }
        },
        "is_valid": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "boolean"
          }
        },
        "latency_ms": {
          "nullable": false,
          "required": true,
          "schema": {
            "minimum": 0.0,
            "type": "integer"
          }
        },
        "question": {
          "nullable": false,
          "required": true,
          "schema": {
            "type": "string"
          }
        },
        "reject_reason": {
          "nullable": true,
          "required": false,
          "schema": {
            "type": "string"
          }
        },
        "row_count": {
          "nullable": false,
          "required": true,
          "schema": {
            "minimum": 0.0,
            "type": "integer"
          }
        },
        "rows": {
          "nullable": false,
          "required": true,
          "schema": {
            "items": {
              "additional_properties": true,
              "fields": {},
              "type": "object",
              "type": "array"
            },
            "type": "object"
          }
        }
      }
    }
  }
}
```

GraphRelationship

```
{
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  "fields": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "relation_id": {
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      "required": true,
      "schema": {
        "pattern": "^REL-[0-9a-f]{20}$",
        "type": "string"
      }
    },
    "to_node_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

GroupedMetricResult

```
{
  "additional_properties": false,
  "fields": {
    "group": {
      "nullable": false,
      "required": true,
      "schema": {
        "additional_properties": true,
        "fields": {},
        "type": "object"
      }
    },
    "value": {
      "nullable": false,
      "required": false,
      "schema": {
        "type": "union",
        "variants": [
          {
            "type": "integer"
          },
          {
            "type": "number"
          },
          {
            "type": "null"
          }
        ]
      }
    }
  },
  "type": "object"
}
```

HTTPValidationError

```
{
  "additional_properties": true,
  "fields": {
    "detail": {
      "nullable": false,
      "required": false,
      "schema": {
        "items": {
          "additional_properties": true,
          "fields": {
            "loc": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "number"
              }
            },
            "type": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            }
          },
          "type": {
            "type": "union",
            "variants": [
              {
                "type": "string"
              },
              {
                "type": "integer"
              }
            ],
            "type": "array"
          },
          "msg": {
            "nullable": false,
            "required": true,
            "schema": {
              "type": "string"
            }
          },
          "type": {
            "nullable": false,
            "required": true,
            "schema": {
              "type": "string"
            }
          }
        },
        "type": "object",
        "type": "array"
      }
    }
  },
  "type": "object"
}
```

HealthResponse

```
{
  "additional_properties": false,
  "fields": {
    "status": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "UP"
        ],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

MetricPlan

```
{
  "additional_properties": false,
  "fields": {
    "column": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "p": {
      "nullable": true,
      "required": false,
      "schema": {
        "maximum": 100.0,
        "minimum": 0.0,
        "type": "number"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "count",
          "max",
          "mean",
          "median",
          "min",
          "percentile",
          "ratio",
          "std",
          "sum"
        ],
        "type": "string"
      }
    }
  },
  "type": "object"
}
```

MetrologyAskEvidence

```
{
  "additional_properties": false,
  "fields": {
    "excerpt": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "source_id": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
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    }
  },
  "type": "object"
}
```

MetrologyEvidence

```
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    },
    "source_id": {
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      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "title": {
      "nullable": false,
      "required": true,
      "schema": {
        "min_length": 1,
        "type": "string"
      }
    },
    "type": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "METROLOGY"
        ],
        "type": "string"
      }
    },
    "rules": [
      "metrology.alarm_result is forbidden"
    ],
    "type": "object"
  }
}
```

NIQueryHistoryResponse

```
{
  "additional_properties": false,
  "fields": {
    "items": {
      "nullable": false,
      "required": true,
      "schema": {
        "items": {
          "additional_properties": false,
          "fields": {
            "asked_at": {
              "nullable": false,
              "required": true,
              "schema": {
                "format": "date-time",
                "type": "string"
              }
            },
            "error_msg": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              }
            },
            "generated_sql": {
              "nullable": true,
              "required": false,
              "schema": {
                "type": "string"
              }
            },
            "is_rejected": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              }
            },
            "is_valid": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "boolean"
              }
            },
            "latency_ms": {
              "nullable": true,
              "required": false,
              "schema": {
                "minimum": 0.0,
                "type": "integer"
              }
            },
            "nl_query_log_id": {
              "nullable": false,
              "required": true,
              "schema": {
                "minimum": 1.0,
                "type": "integer"
              }
            },
            "outcome": {
              "nullable": false,
              "required": true,
              "schema": {
                "enum": [
                  "DB_ERROR",
                  "POLICY_REJECTED",
                  "SUCCESS",
                  "VALIDATION_FAILED"
                ],
                "type": "string"
              }
            },
            "question": {
              "nullable": false,
              "required": true,
              "schema": {
                "type": "string"
              }
            },
            "reject_reason": {
              "nullable": true,
              "required": false,
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            },
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              "nullable": true,
              "required": false,
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                "type": "integer"
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            },
            "type": {
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              "type": "array"
            },
            "page": {
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              "required": true,
              "schema": {
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                "type": "integer"
              }
            },
            "size": {
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              "required": true,
              "schema": {
                "maximum": 100.0,
                "minimum": 1.0,
                "type": "integer"
              }
            },
            "total": {
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              "required": true,
              "schema": {
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                "type": "integer"
              }
            }
          },
          "type": "object"
        },
        "type": "array"
      }
    }
  },
  "type": "object"
}
```

NIQueryLogItem

```
{
  "additional_properties": false,
  "fields": {
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      "required": true,
      "schema": {
        "format": "date-time",
        "type": "string"
      }
    },
    "error_msg": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "generated_sql": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "is_rejected": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    },
    "is_valid": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "boolean"
      }
    },
    "latency_ms": {
      "nullable": true,
      "required": false,
      "schema": {
        "minimum": 0.0,
        "type": "integer"
      }
    },
    "nl_query_log_id": {
      "nullable": false,
      "required": true,
      "schema": {
        "minimum": 1.0,
        "type": "integer"
      }
    },
    "outcome": {
      "nullable": false,
      "required": true,
      "schema": {
        "enum": [
          "DB_ERROR",
          "POLICY_REJECTED",
          "SUCCESS",
          "VALIDATION_FAILED"
        ],
        "type": "string"
      }
    },
    "question": {
      "nullable": false,
      "required": true,
      "schema": {
        "type": "string"
      }
    },
    "reject_reason": {
      "nullable": true,
      "required": false,
      "schema": {
        "type": "string"
      }
    },
    "row_cnt": {
      "nullable": true,
      "required": true,
      "schema": {
        "minimum": 0.0,
        "type": "integer"
      }
    }
  },
  "type": "object"
}
```

```

false, "schema": {"minimum": 0.0, "type": "integer"}}, "type": "object"}
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PublicApprovalDecision
{"enum": ["APPROVED", "REJECTED"], "type": "string"}
PublicApprovalItem
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PublicApprovalStatus
{"enum": ["APPROVED", "PENDING", "REJECTED"], "type": "string"}

```

[illegible]


```
, "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable": false,
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ble": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDENCY_UNAVAILABLE", "KAFKA_L
AG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"}}, "status": {"nullable"
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}, "reason_code": {"nullable": true, "required": true, "schema": {"enum": ["CONTRACT_MISMATCH", "DEPENDEN
CY_UNAVAILABLE", "KAFKA_LAG_STALE", "NOT_CONFIGURED", "RAG_MODEL_NOT_READY", "TIMEOUT"], "type": "string"
}}, "status": {"nullable": false, "required": true, "schema": {"enum": ["FAIL", "PASS"], "type": "string"
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s READY and HTTP 200", "any check FAIL yields status NOT_READY and HTTP 503"], "type": "object"}
RunAlarmEvidence
{"additional_properties": false, "fields": {"alarm": {"nullable": false, "required": true, "schema": {"ad
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DETECTION_COMPLETED / LOT_HIST / A
AGENT_RUN_STARTED / AGENT_RUN / C
HYPOTHESIS_GENERATED / AGENT_RUN / C
APPROVAL_REQUESTED / APPROVAL / C
APPROVAL_DECIDED / APPROVAL / C
ACTION_SENT / ACTION / C delivery service
ACTION_SEND_FAILED / ACTION / C delivery service
AGENT_RUN_COMPLETED / AGENT_RUN / C
AGENT_RUN_FAILED / AGENT_RUN / C